

Executive Report

Adult Income Prediction using Machine Learning

Project Overview

The objective of this project was to develop an end-to-end machine learning solution for predicting whether an individual's annual income exceeds \$50,000 using demographic and employment-related attributes from the UCI Adult Income dataset. The project followed a complete machine learning lifecycle, including data preprocessing, feature engineering, model development, hyperparameter tuning, ensemble learning, interpretability, fairness evaluation, and deployment preparation.

From a business perspective, incorrectly predicting high-income individuals (false positives) is considered more costly than missing some eligible individuals (false negatives). Therefore, **precision** was selected as the primary evaluation metric throughout model development.

Dataset

The Adult Income dataset was obtained from the UCI Machine Learning Repository through OpenML. It contains demographic, educational, occupational, and financial attributes for predicting whether an individual's annual income exceeds \$50,000.

Dataset Summary

- Total observations: **48,842**
- Target variable:
 - $\leq 50K$
 - 50K
- Features:
 - Numerical demographic variables
 - Categorical demographic variables
 - Employment information
 - Capital gain and capital loss
 - Education level
 - Hours worked per week

The dataset was divided into:

- Training set: 70%
- Development set: 10%
- Hold-out test set: 20%

Stratified sampling was used to preserve the original class distribution across all splits.

Feature Engineering

Eight additional features were engineered to improve predictive performance while preventing data leakage.

The engineered features included:

- Age buckets
- Working-hours buckets
- Capital gain indicator
- Capital loss indicator
- Log-transformed capital gain
- Log-transformed capital loss
- Higher education indicator
- Education $\times$ working-hours interaction

These features were incorporated into the preprocessing pipeline using a `FunctionTransformer`, ensuring that the same transformations were automatically applied during both training and inference.

Model Development

Several machine learning models were developed and evaluated throughout the project, including:

- Logistic Regression
- Random Forest
- LightGBM
- Stacking Ensemble (Random Forest + LightGBM with a Logistic Regression meta-learner)

Hyperparameter tuning was performed using `RandomizedSearchCV` with stratified cross-validation. The final LightGBM model was further calibrated using Isotonic Regression to improve probability estimates, which were later used during threshold optimization.

The final deployed model was the calibrated LightGBM pipeline.

Hold-Out Performance

Final Calibrated LightGBM Model

Metric	Value
Accuracy	86.72%
Precision	79.49%
Recall	60.01%
F1-score	68.39%
ROC-AUC	0.9198
PR-AUC	0.7988

Among all evaluated models, the calibrated LightGBM provided the best overall balance between predictive performance and the project's business objectives while maintaining strong precision.

Decision Threshold Selection

Rather than using the default classification threshold of **0.50**, threshold optimization was performed based on the project's business objective.

The selected threshold was:

0.45

This threshold was chosen because it represented the **lowest threshold capable of achieving at least 75% precision**, thereby maximizing recall while still satisfying the required business precision target.

Class Imbalance Evaluation

The training data exhibited moderate class imbalance:

- Negative class ($\leq 50K$): **76.07%**
- Positive class ($> 50K$): **23.93%**

Three imbalance handling techniques were evaluated:

- Class Weighting
- Random Oversampling
- SMOTE

Although these techniques substantially increased recall, they also caused a significant reduction in precision. Since precision was the primary business metric, none of the imbalance handling approaches outperformed the original tuned LightGBM model.

Consequently, the original tuned LightGBM model without additional imbalance handling was retained as the final production model.

Model Interpretability

Model interpretability was evaluated using three complementary approaches:

- LightGBM Feature Importance
- Permutation Importance
- SHAP (SHapley Additive exPlanations)

The most influential features included:

- Capital Gain
- Marital Status
- Education Number
- Capital Loss
- Age
- Education $\times$ Working Hours Interaction
- Executive Manager Occupation

In addition, SHAP waterfall plots were generated for three representative test instances:

- True Positive
- False Positive
- False Negative

These local explanations demonstrated how individual feature values contributed to each prediction, improving model transparency and supporting deployment readiness.

Fairness Assessment

Fairness was assessed by comparing precision across protected demographic groups.

Precision by Sex

Group	Precision
Female	87.96%
Male	82.14%

Precision by Race

Group	Precision
Black	90.28%
White	83.09%
Asian-Pac-Islander	68.06%
Other	100.00%*
Amer-Indian-Eskimo	100.00%*

*These groups contained relatively few samples, so their precision estimates should be interpreted with caution.

Although moderate disparities were observed, particularly for Asian-Pac-Islander individuals, the largest differences occurred in demographic groups with relatively small sample sizes.

Therefore, continued fairness monitoring and evaluation using additional fairness metrics are recommended before production deployment.

Deployment

The final deployment package includes:

- Calibrated LightGBM model
- Saved preprocessing pipeline
- Feature engineering module
- End-to-end inference script
- Input validation
- Unit tests
- Monitoring checklist

The inference pipeline accepts raw input records, automatically performs feature engineering and preprocessing, generates calibrated prediction probabilities, applies the optimized decision threshold of **0.45**, and returns the predicted class along with the three most influential features contributing to the prediction.

Recommendations

The developed machine learning system demonstrates strong predictive performance while remaining interpretable, fair, and deployment-ready.

Future work should focus on:

- Continuously monitoring data quality and model performance.
- Retraining the model approximately every six months or whenever precision falls below the business target.
- Expanding fairness evaluation using additional fairness metrics such as Equal Opportunity and Demographic Parity.
- Conducting A/B testing before full-scale production deployment.
- Evaluating additional gradient boosting algorithms and ensemble techniques on future datasets.

Overall, this project successfully delivered a complete end-to-end machine learning pipeline that satisfies the stated business objective while incorporating best practices in model evaluation, threshold optimization, interpretability, fairness assessment, deployment, and monitoring.